

**Title:” Hybrid AI Driven Traffic Energy
Harvesting for Smart Cities”**

Abstract:

- The rapid increase in vehicular traffic has led to significant energy dissipation in the form of mechanical vibrations, heat, and friction. Harnessing this wasted energy can contribute to sustainable power generation. This project proposes a **hybrid AI-driven traffic energy harvesting system** that integrates multiple technologies—piezoelectric, electromagnetic, triboelectric, thermoelectric, and solar energy harvesting. Artificial Intelligence (AI) is employed to optimize energy prediction, manage hybrid sources, and improve efficiency based on traffic density and environmental conditions. The system aims to provide a scalable, eco-friendly solution for powering roadside infrastructure such as streetlights, traffic signals, and IoT devices

Keywords:

- Hybrid Energy Harvesting
- Artificial Intelligence
- Piezoelectric
- Electromagnetic
- Triboelectric
- Thermoelectric
- Solar Energy
- Smart Traffic System
- Sustainable Power

Introduction:

- Urbanization and the rise in vehicle usage have created challenges in energy demand and environmental sustainability. Roads and highways experience constant mechanical stress, vibrations, and heat dissipation due to traffic flow. These wasted energies can be harvested using advanced technologies. Traditional single-source energy harvesting systems often suffer from low efficiency and limited scalability.
- A **hybrid approach**, combining multiple energy harvesting technologies, ensures higher energy output and reliability. By integrating AI algorithms, the system can predict traffic flow, optimize energy conversion, and intelligently distribute harvested energy. This makes the solution suitable for smart cities and sustainable infrastructure development.

Existing System:

- Current traffic energy harvesting systems typically rely on single-source technologies due to that low energy emission
- High noise pollution due the traffic congestion.
- Overcrowding which leads accident during traffics.
- Lack of intelligent control and prediction..
- Dependent on power grids
- High pollutant emission without any energy recovery

Proposed System:

- Piezoelectric system: Convert mechanical stress from vehicle movement into electrical energy.
- Electromagnetic systems: Use coils and magnets to generate electricity from the vehicles.
- Triboelectric systems: Harvest energy from friction between surfaces.
- Thermoelectric systems: Capture heat from vehicle engines and road surfaces.
- Solar panels: Installed near roads to capture sunlight.

1. Piezoelectric Sensors

- **Placement:** Embedded directly under the road surface, especially at high-traffic zones like intersections, toll booths, and speed breakers.
- **Reason:** Vehicle pressure and vibrations are strongest here, ensuring maximum energy capture.

2. Solar Panels

- **Placement:** Mounted on streetlights, traffic signals, rooftops of nearby buildings, or noise barriers along highways.
- **Reason:** Elevated positions ensure maximum sunlight exposure without obstruction.
- **Output Use:** Daytime energy supply for lighting, signage, and charging stations.

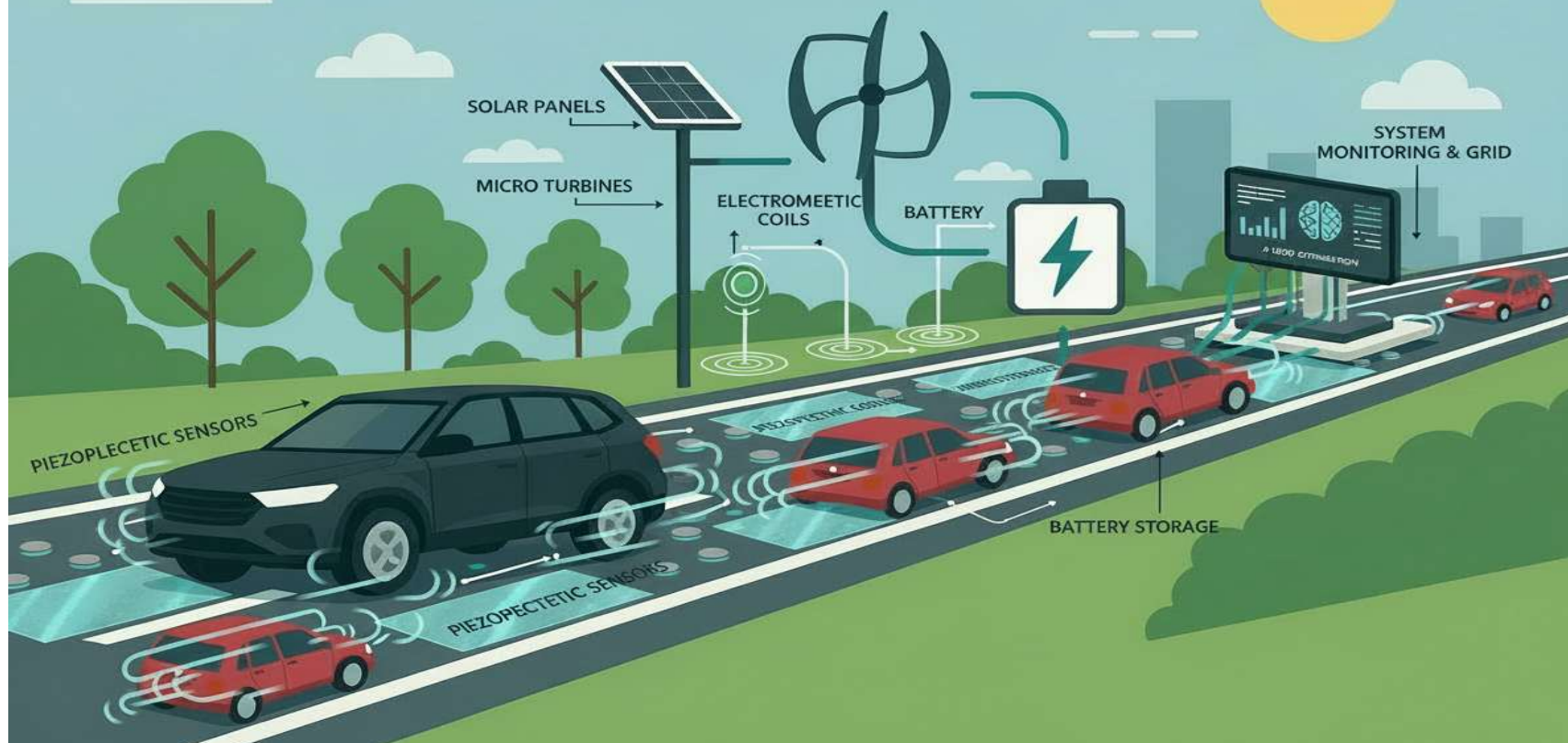
3. Microturbines

- **Placement:** Along the roadside, near barriers, or on poles where airflow from moving vehicles is strongest.
- Can also be installed in **rainwater downpipes** or **bridge structures** to capture water flow or vibrations.
- **Reason:** Vehicle slipstreams and natural wind provide continuous airflow; water flow in urban drainage adds another source.
- **Output Use:** Supplementary power for sensors, IoT devices, or feeding into the grid.

4. Electromagnetic Coils (if included)

- **Placement:** Beneath speed bumps or modular tiles.
- **Reason:** Converts vertical displacement and motion into electricity.
- **Output Use:** Ideal for powering nearby monitoring systems.

DYNAMIC TRAFFIC ENERGY HARVESTING



Conclusion:

- Hybrid ai driven traffic energy harvesting for smart cities provide a social well-being long -term environment. It uses renewable energy resources , ai optimization to predicate traffic flow and environmental conditions. And prevent energy wastage , emission of pollutants and enhance smart urban infrastructure